RESEARCH PROJECT & PROPOSAL APPROVAL BRIEF

Cascaded Adaptive Blended RAG: Optimizing Retrieval Efficiency via Pre-Retrieval Query Cascading

Author: Master's Student **Target Base Paper:** Sawarkar et al. (2024) **Focus:** RAG Efficiency & Adaptive Routing

1. Base Paper Selection & Motivation

Selected Base Paper: *'Blended RAG: Improving RAG (Retriever-Augmented Generation) Accuracy with Semantic Search and Hybrid Query-Based Retrievers'* (Sawarkar et al., 2024, IEEE MIPR).

Why This Paper? The overall accuracy of Retrieval-Augmented Generation (RAG) systems heavily depends on retriever quality. Sawarkar et al. demonstrated that combining dense vector indexes and sparse encoder indexes with multi-field hybrid query strategies (such as *Best Fields* search across titles, abstracts, and body contexts) significantly outperforms single-retriever baselines (e.g., standard dense embeddings or BM25 keyword search) and avoids costly domain-specific model fine-tuning.

THE CORE PROBLEM & RESEARCH GAP: THE STATIC HYBRID PENALTY

While Blended RAG achieves top-tier retrieval accuracy across benchmarks like NQ and HotpotQA, it suffers from a major computational bottleneck: **The Static Hybrid Penalty**. Blended RAG executes full multi-field hybrid queries (Dense Vector Search + Sparse ELSER Search + Multi-Field Expansion) for every incoming query regardless of difficulty. Simple factual queries that could easily be answered by a fast lexical lookup pay the exact same latency and compute cost as complex compositional queries.

2. Proposed Research Extension: Cascaded Adaptive Blended RAG

CORE ARCHITECTURAL OBJECTIVE

Design a lightweight **Pre-Retrieval Cascaded Router** that evaluates query specificity first. Simple factual queries exit early via a fast BM25 pass, while complex queries escalate to full Blended RAG (BM25 + SPLADE + Dense MiniLM).

QUANTITATIVE EFFICIENCY TARGET

Reduce average query latency and neural vector search invocations by **40–60%** while preserving **≥ 95%** of full static Blended RAG retrieval accuracy.

Pre-Retrieval Feature Analysis — STEP 1 · TIER 0

Extracts zero-overhead lexical specificity signals directly from the query text (token count, rare token ratio, maximum Inverse Document Frequency $MaxIDF$) prior to index lookup.

BM25 Fast Pass & Early Exit — STEP 2 · TIER 1

Executes a fast lexical lookup. If top-1 score $s_1 \geq \tau_{\text{high}}$ or score margin $\Delta = s_1 - s_2 \geq \tau_{\text{gap}}$ is wide, returns BM25 results immediately and skips neural search.

Fallback 3-Way Blended Search — STEP 3 · TIER 2

When lexical confidence is low ($\Delta < \tau_{\text{gap}}$), escalates to full hybrid search combining BM25, SPLADE Sparse Encoder, and Dense MiniLM via Reciprocal Rank Fusion (RRF).

Multi-Field Metadata Matching — CROSS-TIER RULE

Applies *Best Fields* and *Cross Fields* matching across structured document fields (`title`, `abstract`, body), preserving Blended RAG's core accuracy advantage.

3. Research Novelty & System Architecture

HOW OUR CASCADE DIFFERS FROM RECENT ADAPTIVE HYBRID RAG LITERATURE (2025–2026)

PRIOR WORK LIMITATION · POST-RETRIEVAL WEIGHTING

DAT (Hsu et al., 2025) & QuDAR (Kim et al.)

Adjust the mixing weight (α) between BM25 and Dense scores per query, but must execute **both sparse and dense retrievers on 100% of queries first**—saving **0%** execution latency.

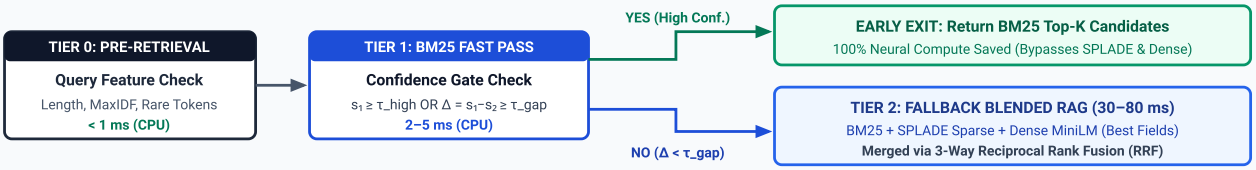
LATENCY VS. ACCURACY POSITIONING

OUR CONTRIBUTION · PRE-RETRIEVAL EARLY EXIT

Cascaded Adaptive Blended RAG

Evaluates query specificity and BM25 margin (Δ) **before** invoking neural models—halting at Tier 1 on unambiguous queries to cut **40–60%** of vector search compute.

SEQUENTIAL PRE-RETRIEVAL & LEXICAL CASCADE EXECUTION FLOW



PIPELINE STAGE	MECHANISM & MODELS USED	COMPUTE COST	TARGET QUERY TYPE
Tier 0: Pre-Retrieval	Query Length, IDF Statistics, Rare Tokens	< 1 ms (CPU)	All incoming user queries
Tier 1: BM25 Fast Pass	BM25 Lexical Index + Margin Check (Δ)	2–5 ms (CPU)	Simple factual / exact-keyword match
Tier 2: Blended Hybrid	BM25 + SPLADE Sparse + Dense MiniLM + RRF	30–80 ms (CPU)	Complex, compositional, or abstract
Field Strategy	*Best Fields* / *Cross Fields* on Title & Abstract	Zero added cost	Metadata-rich corpora (e.g., SciFact)

4. Experimental Setup & Evaluation Protocol

- **Datasets & Environment:** Evaluated on CPU-friendly hardware using BEIR **SciFact** (5,183 scientific documents with `title` and `abstract` metadata fields, 300 test queries) and a reproducible subset of **Natural Questions (NQ)**.
- **Baselines Compared:** (1) Standard BM25, (2) Dense Vector Search (`all-MiniLM-L6-v2`), (3) Static Blended RAG (Sawarkar et al., evaluating both single-index Sparse Encoder + *Best Fields* and 3-way Multi-Field RRF), and (4) **Cascaded Adaptive Blended RAG (Proposed)**.
- **Evaluation Metrics & Targets: Retrieval Quality** (Recall@10, MRR@10, nDCG@10 — maintain ≥ 95% of static Blended RAG); **Efficiency** (ms/query, % neural calls avoided — reduce latency by 40–60%); and **Downstream Generation** (Exact Match & F1 via local `Llama-3.2-3B`).